

● RAG Learning

Eduqas GCSE Geography B (C112QS) — Theme 3: Environmental Challenges

3.1 How Ecosystems Function

1. Biomes and Global Climate Zones

A **biome** is a very large scale ecosystem, defined by its climate and the plant and animal communities that climate supports. Biomes are not scattered at random: their distribution is controlled almost entirely by the **global circulation of the atmosphere**, which fixes where air rises (low pressure, wet) and where it sinks (high pressure, dry). Because that circulation is arranged in belts either side of the equator, the world's biomes also form broad latitudinal belts.

Biome: A global scale ecosystem in which the climate, soils, vegetation and animal life are closely interdependent — for example tropical rainforest, hot semi-arid grassland (savanna), tundra or temperate deciduous forest.

How Global Circulation Fixes Biome Distribution

Latitude	Air movement	Pressure and rainfall	Resulting biome
0° (equator)	Rising air at the ITCZ	Low pressure; rain all year, over 2,000 mm	Tropical rainforest
5–20° N and S	ITCZ migrates north and south with the overhead sun	Alternating low and high pressure; a wet season and a long dry season	Hot semi-arid grassland (savanna)
20–30° N and S	Descending limb of the Hadley cell	Permanent high pressure; under 250 mm	Hot desert
40–60° N	Rising air where polar and tropical air meet	Low pressure; rain spread through the year	Temperate deciduous forest

60–70° N	Cold, dry, sinking air	High pressure; low precipitation, mostly snow	Tundra
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The Eduqas specification requires you to know **two contrasting biomes in detail**, one of which **must be hot semi-arid grassland**. These notes use hot semi-arid grassland (the savanna) and tropical rainforest, which sit next to each other on the globe but function very differently — a contrast that makes comparison questions far easier to answer.

Exam Tip — Describing Distribution: When asked to describe where a biome is found, always name the latitude band, name two or three specific regions or countries, and link the pattern to pressure. "Savanna is found roughly between 5° and 20° north and south of the equator — across the Sahel, East Africa, northern Australia and the Brazilian cerrado — where the ITCZ brings rain for part of the year and high pressure dominates for the rest." That single sentence scores far more than "it is found in hot places".

2. Hot Semi-Arid Grassland (Savanna)

Hot semi-arid grassland — usually called **savanna** — is the compulsory biome for this specification. It is a transition biome, sitting between the rainforest at the equator and the hot deserts at around 25°. That in-between position is the key to everything about it: it receives enough rain to support grass but not enough to support closed forest, and the rain arrives in one concentrated season rather than all year.

Location and Distribution

- **Africa:** The largest continuous area — the Sahel belt south of the Sahara, and the East African savannas of Kenya and Tanzania (the Serengeti and Maasai Mara).
- **South America:** The Brazilian *cerrado* and the Venezuelan *llanos*.
- **Australia:** The tropical north, across the Northern Territory and Queensland.
- **Asia:** Parts of central India and Myanmar.

Climate Characteristics

Variable	Typical value	Detail
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Mean annual temperature	25–30°C	Hot all year; the hottest months come just before the rains break, when there is no cloud cover.
Annual rainfall	Roughly 500–1,500 mm	Concentrated into a single wet season; the Serengeti receives around 900–1,000 mm across much of the park, falling below 800 mm in the drier south-east.
Dry season length	Typically 5–7 months	Can be as short as 2 months at the wetter margin and as long as 11 months at the desert margin.
Rainfall reliability	Low and falling	Totals vary sharply from year to year — the defining hazard of the biome and the reason it is vulnerable to desertification (see 3.4).

The wet season arrives when the **ITCZ migrates** towards the savanna as the overhead sun moves north (June–August in the northern hemisphere) or south (December–February in the southern hemisphere). For the rest of the year the descending, warming, drying air of the **subtropical high pressure belt** sits over the region, and almost no rain falls.

Soils

Savanna soils are typically reddish-brown **ferruginous (iron-rich) soils**. They differ from rainforest soils in two important ways:

- **More nutrients are held in the soil itself.** Rainfall is not heavy enough all year to leach nutrients deep out of reach, so a thin but real nutrient store builds up in the topsoil.
- **They form a hard crust.** In the dry season the surface bakes and can develop a *laterite* hardpan. When the first heavy rains arrive the water cannot infiltrate, so it runs off, and gullies form. This is why the start of the wet season — not the dry season — is when savanna soil erosion is worst.

Vegetation

Savanna vegetation is a **continuous grass layer with scattered, widely spaced trees**. Everything about it is a response to the long dry season and to fire.

Adaptation	Example	How it helps survival
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Deep tap roots	Acacia — roots reaching well beyond the depth of the grasses	Reaches groundwater that the grasses cannot, so the tree survives months without rain
Water storage in the trunk	Baobab	The swollen trunk stores water drawn up during the wet season and drawn on in the dry
Deciduous in the dry season	Most savanna trees	Dropping leaves cuts transpiration to almost nothing when water is scarce
Small, waxy or thorn-like leaves	Acacia	Reduces the surface area losing water; thorns also deter browsing by giraffe and antelope
Umbrella-shaped crown	Acacia	Shades the ground directly beneath the tree, cutting evaporation from the soil around the roots
Fire-resistant thick bark	Many savanna trees	Dry season fires are a normal part of the ecosystem; thick bark protects the living tissue underneath
Grasses grow from the base	Red oat grass, elephant grass	The growing point sits at or below ground level, so grazing and fire remove the leaf but not the plant, and regrowth is rapid after rain

Animals

Savanna animals are adapted to seasonal, unreliable food and water and to open ground with few hiding places.

- **Migration:** Around 1.3 million wildebeest, along with zebra and gazelle, move in an annual circuit between the Serengeti and the Maasai Mara, following the rains and the fresh grass. Migration is the single most important adaptation in the biome — the animals move to the resource rather than waiting for it.
- **Grazing succession:** Zebra eat the tough upper stems first, wildebeest then take the middle leaf, and gazelle crop the short new shoots beneath. Three species use the same grass without competing — a clear example of interdependence.
- **Water conservation:** The oryx and the gerenuk can obtain most of their water from their food and tolerate a rise in body temperature rather than sweating it away.

- **Speed and camouflage:** With no cover, prey animals rely on speed and herd vigilance; predators such as the cheetah and lion rely on short bursts of speed and on dappled or tawny coats matching dry grass.
- **Termites:** Termite mounds are the biome's engine room — termites break down dry, tough grass litter that few other organisms can digest, returning nutrients to the soil, and their mounds create drainage channels and raised, nutrient-rich patches that trees colonise.

People

Humans are part of the savanna ecosystem, not separate from it. Traditional pastoralists such as the **Maasai** of Kenya and Tanzania and the **Fulani** of the Sahel move cattle, goats and sheep between seasonal grazing areas and water points — a mobile system that matches the mobility of the wild herds and lets grazed land recover. Where movement is restricted by fences, national park boundaries or land sales, the same number of animals is concentrated on less land, grazing pressure rises and the vegetation cover thins. That is the starting point for the damage covered in 3.2 and for desertification in 3.4.

Case Study — The Serengeti–Mara Ecosystem, Tanzania and Kenya

Climate: Roughly 900–1,000 mm of rain a year across much of the Serengeti, dropping below 800 mm in the drier south-east and rising to around 1,400 mm in the north. Rain comes in two pulses — short rains from October to December and long rains from March to May — separated by a pronounced dry season from June to September.

Soils: In the south-east, volcanic ash from the Ngorongoro highlands has produced shallow, calcium-rich soils that support short, highly nutritious grass. This is why the wildebeest calve there.

Vegetation: Short-grass plains in the south-east grade into taller grassland and then into Acacia–Commiphora woodland in the north and west.

Interdependence in action: The grass feeds the migrating herds; the herds' dung and urine return nitrogen to the soil; their grazing keeps the grass short enough to stop woody scrub taking over; the herds feed lion, hyena and crocodile; and dry season fires, set by lightning and by people, clear dead material and release nutrients. Remove any one link and the plains change character.

Exam Tip — Interdependence: Eduqas asks specifically about the *interdependence* of climate, soils, vegetation, animals and humans. The strongest answers run a chain: seasonal rainfall → grasses that grow from the base → grazing herds that can crop them without killing them → dung returning

nutrients → soil fertility maintained → grasses grow again. Then break the chain and say what happens. Listing the five components separately scores in the bottom band.

3. Tropical Rainforest — Location and Climate

Tropical rainforests are found between the **Tropic of Cancer (23.5°N)** and **Tropic of Capricorn (23.5°S)**, with the densest concentration within 5° either side of the equator. Their location is driven by the **Inter-Tropical Convergence Zone (ITCZ)** — a belt of low pressure where trade winds meet, causing warm, moist air to rise, cool, and produce intense rainfall.

Climate Characteristics

Variable	Value	Detail
Mean annual temperature	~27°C	Very low seasonal variation (1–2°C). Hot all year due to direct overhead sun.
Annual rainfall	>2,000 mm/year	Rainfall distributed throughout the year; convectional rainfall common in afternoons.
Humidity	Very high (80–90%)	Constant evapotranspiration from dense vegetation keeps air saturated.
Seasonality	None / very slight	No true dry season at the equator; ITCZ movement creates slight wetter/drier periods.

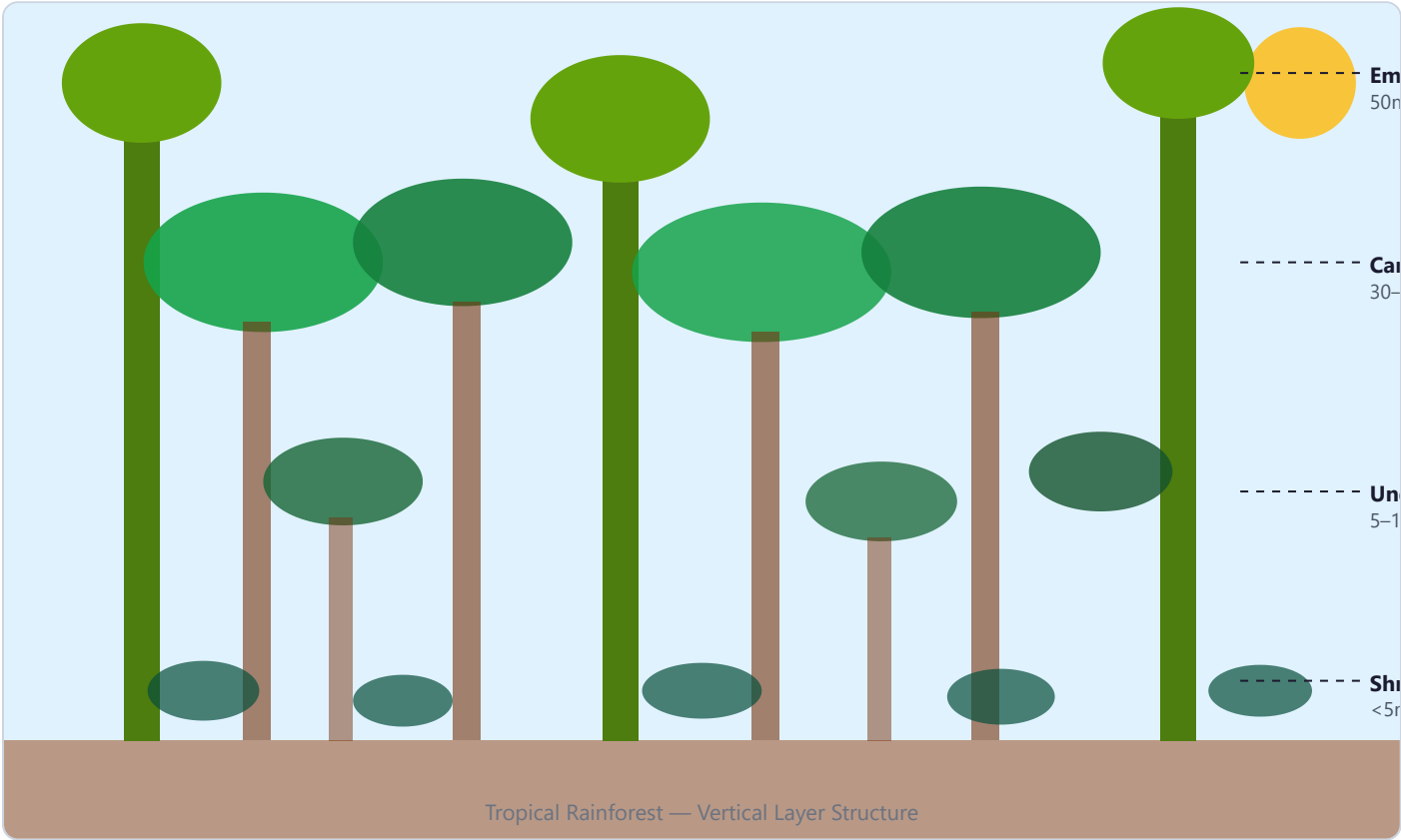
Major Tropical Rainforest Areas

Region	Location	Extent
Amazon Basin	South America (mainly Brazil)	~5.5 million km ² — largest in the world
Congo Basin	Central Africa (DRC, Congo, Cameroon)	~1.7 million km ² — second largest

Borneo / SE Asian forests	Indonesia, Malaysia, Papua New Guinea	Highly threatened; orangutan habitat
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4. Structure of the Rainforest

The rainforest has a **stratified structure** — four distinct vertical layers, each with different light levels, temperature, and wildlife. This layering creates dozens of different microhabitats supporting extraordinary biodiversity.



Layer	Height	Light Level	Plants	Animals
Emergent	50m+	Full sunlight	Kapok, Brazil nut tree — isolated giant trees with umbrella-shaped crowns	Harpy eagles, macaws, butterflies

Canopy	30–40m	~80% of light	Dense, interlocking branches forming a continuous roof; where 80% of photosynthesis occurs	Monkeys, toucans, tree frogs, sloths
Understorey	5–15m	Very low (~2–5%)	Adapted to shade; large leaves to maximise light absorption; young canopy trees	Jaguars, snakes, insects
Shrub/Ground	<5m	Less than 1%	Sparse; mosses, ferns, fungi, seedlings that wait for a gap in canopy	Tapirs, ants, decomposers, gorillas (Congo)

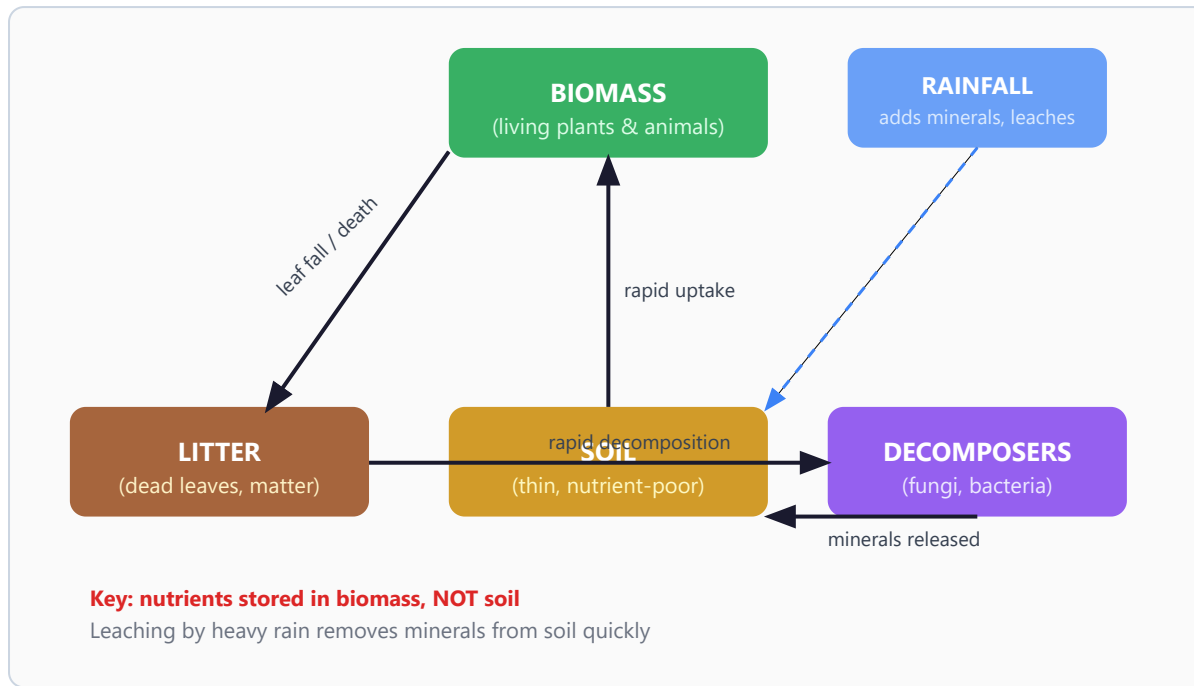
Exam Tip: When describing rainforest structure, always link the layer to its light level and explain how plants in that layer are adapted to survive with that amount of light. This shows geographical understanding rather than just recall.

5. Nutrient Cycling

The tropical rainforest has a **rapid nutrient cycle**, driven by the hot, wet climate. Despite the lush vegetation, the soils are actually **thin and nutrient-poor (latosols/oxisols)** — almost all nutrients are locked up in living biomass (plants and animals).

How the Nutrient Cycle Works

Dead leaves and plant matter (litter) fall to the forest floor. The warm, moist conditions mean **decomposers (fungi and bacteria) break down this litter extremely quickly** — sometimes within weeks. Nutrients are released into the thin topsoil and immediately taken up by the shallow root systems of trees. The cycle is therefore almost closed — nutrients flow rapidly from litter to decomposers to soil to plants and back again.



Additional Nuance: The Role of Mycorrhizal Networks

A critical but often overlooked component of the nutrient cycle is the **mycorrhizal fungal network** — a web of fungal threads (hyphae) that connects the roots of different trees underground. These fungi form mutually beneficial (mutualistic) relationships with tree roots: the tree provides the fungus with sugars produced by photosynthesis, and in return the fungus dramatically extends the tree's ability to absorb phosphorus, nitrogen, and water from the soil. This means nutrients can be transferred between trees through the fungal network, making the cycle even more efficient and resilient. In old-growth rainforest, up to 90% of nutrient uptake may be mediated by mycorrhizal fungi.

Why Leaching is Such a Threat

The intense rainfall (2,000+ mm/year) constantly percolates through the soil, dissolving and washing nutrients downward in a process called **leaching**. This is why the nutrient cycle must be so fast — any nutrients that are not taken up almost immediately by roots or fungi are lost to deeper layers and the water table. In intact forest, the canopy intercepts much of the rainfall and roots recapture leached nutrients before they are permanently lost. When the forest is cleared, this safety net disappears entirely.

Buttress Roots: Large, plank-like roots that extend from the base of tall trees. They spread the enormous weight of the tree over a wider area (since soils are shallow and unable to support deep

taproots) and also allow the tree to access nutrients near the surface, where decomposition deposits them.

Laterisation: The process by which, following deforestation, the hot sun bakes the exposed, nutrient-depleted soil into a hard, iron-rich crust (laterite). This crust is almost impermeable to water and impossible to cultivate. It can form within a few years of forest clearance and is essentially irreversible on human timescales.

Exam Tip: A key misunderstanding is that rainforest soils are fertile. They are NOT — they are thin and poor. If the forest is cleared, heavy rain leaches away remaining nutrients quickly (laterisation), leaving hard, brick-like soil useless for farming. This is why slash-and-burn farmers must move on after only 2–3 years — the soils are exhausted almost immediately.

Exam Tip — Nutrient Cycle Comparisons: If asked to compare the rainforest nutrient cycle to a temperate deciduous forest, the key differences are: (1) speed — decomposition is far faster in the tropics due to heat and moisture; (2) storage — almost all nutrients are in biomass in the rainforest, whereas temperate soils hold more nutrients; (3) leaching risk — far greater in the rainforest due to higher rainfall intensity.

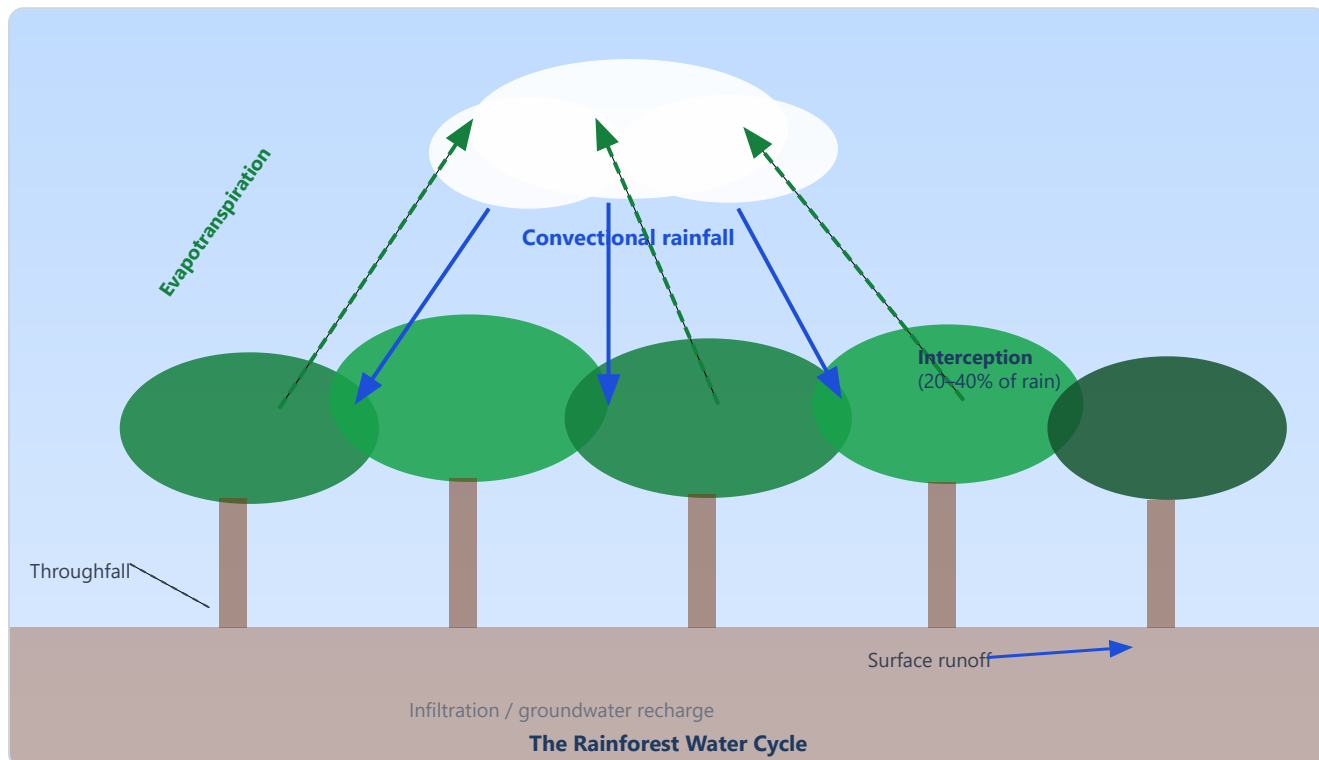
6. The Water Cycle in the Rainforest

The tropical rainforest plays an extraordinary role in the local and global water cycle. The forest does not simply receive rainfall — it actively **generates its own rainfall** through evapotranspiration, creating a self-sustaining feedback loop between vegetation and climate.

Key Processes

Interception: The process by which rainfall is caught by the leaves, branches, and bark of vegetation before it reaches the ground. In a tropical rainforest, the dense multi-layer canopy can intercept 20–40% of all precipitation. This water either evaporates back into the atmosphere directly (interception loss) or drips slowly to the ground as throughfall, greatly reducing the erosive impact of heavy rain on the soil surface.

Evapotranspiration (EVT): The combined process of evaporation (water evaporating from soil and leaf surfaces) and transpiration (water released through the stomata of leaves as a by-product of photosynthesis). In the Amazon, trees release approximately 20 billion tonnes of water vapour into the atmosphere every single day — more than the Amazon River discharges into the Atlantic Ocean. This moisture rises, cools, condenses, and falls again as rainfall — a process sometimes called the "flying rivers" or "forest-rainfall feedback."



The Forest-Rainfall Feedback Loop

One of the most important — and alarming — aspects of the rainforest water cycle is the **forest-rainfall feedback**. The Amazon generates roughly half of its own rainfall internally through evapotranspiration. Moisture evaporated in the eastern Amazon is carried westward by the trade winds, condenses, falls as rain further west, evaporates again, and is carried still further — creating a "flying river" of moisture that travels thousands of kilometres inland. This means:

- The rainforest is not simply responding to climate — it is **creating the climate it needs to survive**.
- Deforestation at a regional scale interrupts this cycle. Less forest means less evapotranspiration, meaning less moisture in the atmosphere, meaning less rainfall — meaning even more forest dies in a **positive feedback (vicious cycle)**.

- Research by Antonio Nobre (INPA) suggests that the Amazon generates "flying rivers" carrying up to 20 billion tonnes of water per day westward — more than the Amazon River itself. Cities like São Paulo receive much of their water from this invisible aerial river.

Interception in Detail

The dense multi-layered canopy acts as a giant sponge, cushioning the impact of intense tropical rainfall.

Key effects include:

- **Reduced soil erosion:** Rain drops are slowed by leaves and branches before reaching the forest floor, reducing their kinetic energy and the ability to dislodge soil particles.
- **Regulated river flow:** Water reaches rivers more slowly (as throughfall and stemflow rather than direct surface runoff), reducing flood risk downstream.
- **Moisture recycling:** Intercepted water that evaporates directly from leaf surfaces (without entering the soil) feeds local cloud formation and convectional rainfall — keeping the water cycle local and continuous.

Exam Tip — Water Cycle Links to Deforestation: When asked about the effects of deforestation, always include the impact on the water cycle. Without tree cover: (1) interception is lost so rainfall hits bare soil directly, causing erosion and surface runoff; (2) evapotranspiration falls, reducing moisture in the atmosphere and regional rainfall; (3) rivers flood more frequently and more severely. This also links directly to soil degradation and the nutrient cycle.

7. Energy Flows and Food Webs

Nutrient cycling moves matter round and round an ecosystem. **Energy does not cycle** — it enters as sunlight, passes along a one-way chain, and is lost as heat. Understanding that difference is the point of this part of the specification.

Producer: A green plant that converts sunlight into chemical energy by photosynthesis. Producers form the base of every food web.

Consumer: An animal that obtains energy by eating other organisms — primary consumers eat plants, secondary consumers eat primary consumers, tertiary consumers eat secondary consumers.

Decomposer: A bacterium, fungus or detritivore that breaks down dead material and returns nutrients to the soil.

Trophic Levels

Each step in a food chain is a **trophic level**. Only about **10% of the energy** at one level is passed on to the next — the rest is used in respiration, movement and body heat, or is lost in waste. This is why food chains rarely have more than four or five links, and why there are always far fewer top predators than producers.

Trophic level	Tropical rainforest	Hot semi-arid grassland
Producers	Canopy trees, lianas, epiphytes, forest-floor ferns	Red oat grass, elephant grass, acacia, baobab
Primary consumers	Leaf-cutter ants, sloths, howler monkeys, tapir	Wildebeest, zebra, gazelle, elephant, termites
Secondary consumers	Poison dart frogs, tree snakes, army ants	Jackal, serval, secretary bird
Tertiary consumers	Jaguar, harpy eagle, anaconda	Lion, cheetah, spotted hyena
Decomposers	Fungi, bacteria, termites — working within days in the constant warmth and damp	Termites and dung beetles in the wet season; decomposition almost stops in the dry season

Why the Two Biomes Differ

- **Rainforest — energy concentrated at the base.** Constant warmth, light and water mean net primary productivity is the highest of any land biome. Most of the biomass sits in the standing trees, and most consumers are small-bodied specialists living in the canopy.
- **Savanna — energy concentrated in large herbivores.** Productivity is lower and strongly seasonal, but grass is far easier to eat than wood, so an unusually large share of the plant growth is converted into herbivore biomass. This is why the savanna, not the rainforest, carries the world's great herds.
- **Decomposition speed.** In the rainforest, leaf litter is broken down in weeks and nutrients are taken straight back up by shallow roots. In the savanna, dead grass lies dry and unrotted for months, which is

why fire, rather than decomposition, is often what recycles it.

Biodiversity and Food Web Stability

A food web with many species at each level is more **resilient**: if one prey species declines, predators can switch to another, and the web holds. A simplified web — for example one created when farming replaces natural vegetation with a single crop — has no such alternatives, so the loss of one species can collapse several others. This is the mechanism behind the damage described in 3.2.

Exam Tip — Drawing and Reading Food Webs: Arrows in a food web always point in the direction that *energy flows*, i.e. from the organism being eaten to the organism eating it. A common and costly error is to draw them the other way. If asked to predict the effect of removing a species, trace every arrow that touches it and describe both the level above (loses a food source) and the level below (population rises).

8. Biodiversity

The tropical rainforest is the **most biodiverse biome on Earth**, home to over 50% of the world's species while covering only about 6% of the land surface.

Why is Biodiversity so High?

- **Stable climate:** Constant warmth and rainfall all year round allows year-round growth and reproduction — no harsh winter or dry season to kill off species.
- **Multiple niches:** The layered structure creates dozens of distinct microhabitats — different species can specialise without competing directly.
- **Long evolutionary time:** Rainforests have existed for ~60–100 million years without major ice-age disruption, allowing slow accumulation of species through evolution and speciation.
- **High productivity:** Intense photosynthesis produces vast amounts of food energy supporting complex food webs.

Interdependence

Species in the rainforest are **highly interdependent** — the loss of one species can trigger cascading effects throughout the ecosystem. This is sometimes called an ecological cascade or trophic cascade. Key

examples:

- **Brazil nut tree (*Bertholletia excelsa*) — a three-species dependency:** The Brazil nut tree can only be pollinated by the large-bodied **orchid bee** (genus *Eulaema*), because only this bee is strong enough to lift the heavy petal lid of the flower. The orchid bee itself depends on specific orchid species for a chemical it uses in mating displays — meaning no orchids equals no bees equals no Brazil nuts. Once pollinated, the hard woody seed pod can only be opened by the **agouti**, a large rodent with teeth strong enough to gnaw through the shell. The agouti buries seeds it does not immediately eat — and many of these are never retrieved, becoming new trees. Remove any one of these three species and the whole chain collapses.
- **Fig trees and fig wasps:** Every species of fig tree (over 750 species globally) is pollinated by one or a very small number of dedicated wasp species. The wasp lays eggs inside the fig, the young wasps disperse pollen as they leave. The relationship is so tight that extinction of the wasp means extinction of the fig, and with it the loss of a key food source for dozens of bird and mammal species that depend on fig fruits.
- **Jaguars as apex predators (trophic cascade):** The jaguar controls populations of herbivores including peccaries (wild pigs), capybaras, and deer. Where jaguars have been removed, herbivore populations explode, leading to overgrazing of understorey vegetation and a collapse in plant diversity. This then affects insects, birds, and other species that depend on those plants — demonstrating how removing a single apex predator can restructure an entire ecosystem.
- **Epiphytes and pond-in-the-canopy ecosystems:** Bromeliads (relatives of the pineapple) collect rainwater in their leaf rosettes, forming miniature ponds up to 3 litres in volume high in the canopy. These tiny pools support entire independent ecosystems — specialised mosquito larvae, aquatic beetles, poison dart frog tadpoles, and bacteria found nowhere else. A single large bromeliad may support over 250 animal species. This illustrates how one plant species can create an entire new habitat layer.
- **Dung beetles and forest regeneration:** Dung beetles bury animal droppings in the soil, dispersing seeds and aerating the soil. Research in the Amazon showed that when large mammals (which produce dung) are removed by hunting, dung beetle populations collapse, seed dispersal is drastically reduced, and forest regeneration slows significantly. This is an invisible form of biodiversity collapse.
- **Strangler figs and forest succession:** Strangler figs germinate high in the canopy as epiphytes, then slowly send roots down to the ground, eventually engulfing and killing the host tree. The dead host leaves a hollow space that becomes critical shelter for bats, insects, and small mammals — turning a process of apparent destruction into habitat creation.

Exam Tip — Interdependence in Extended Answers: For 6–8 mark questions on biodiversity, do not just list species. Pick one or two specific interdependence examples (e.g. the Brazil nut tree chain, or the jaguar trophic cascade) and explain the mechanism clearly: what depends on what, and what happens if the chain is broken. This demonstrates the depth of understanding that separates high from low band answers.

9. Plant Adaptations

Adaptation	What it Does	Why it Matters
Drip tips	Long, pointed leaf tips	Channels rain off the leaf quickly, preventing the growth of mosses and algae that block sunlight and weigh down leaves
Waxy/smooth leaves	Hydrophobic leaf surface	Repels excess water, preventing fungal/bacterial disease in the constantly wet environment
Large leaves	Maximise surface area	Allows understorey plants to capture the tiny amount of light that penetrates the canopy
Buttress roots	Wide plank-like roots at base	Support enormous tree height in shallow soils; access surface nutrients
Epiphytes	Grow on other plants (not parasitic)	Orchids and bromeliads use tree branches to access light in the canopy without needing ground soil
Lianas	Woody climbing vines	Use trees as support structures to reach canopy light without investing energy in a thick woody trunk
Shallow roots	Roots spread wide near surface	Captures nutrients quickly before they are leached away by rain

Epiphyte: A plant that grows on another plant for physical support but is not parasitic — it obtains its own water (from rain and air moisture) and nutrients (from debris). Examples: orchids, bromeliads, mosses, ferns. Found especially in the canopy layer.

Liana: A woody climbing vine rooted in the soil that uses other trees for support to climb to the canopy. Can grow to over 100m long. Important food source and highway for canopy animals.

10. Animal Adaptations

Animal	Adaptation	Purpose
Toucan	Long, serrated beak (up to 20cm)	Reaches fruit on thin branches that cannot support the bird's weight; also used in social display
Red-eyed tree frog	Adhesive toe pads (suckers); bulging eyes	Grip smooth, wet leaves/branches in canopy; large eyes for low-light understorey conditions
Jaguar	Spotted coat (dappled pattern)	Camouflage in the broken light of the forest floor; ambush predator
Sloth	Extremely slow movement; low metabolic rate; stomach bacteria to digest leaves	Leaves have very low nutritional value; slow movement conserves energy and makes the sloth hard for predators to spot
Poison dart frog	Bright warning colouration (aposematism); skin toxins derived from their diet of ants/mites	Warns predators of toxicity; the toxins themselves come from their diet of specific insects found only in the rainforest
Flying squirrel / gliding animals	Skin membrane between limbs	Glides between trees — efficient travel in the dense forest canopy without descending to the dangerous forest floor

11. Ecosystem Services and Climate Regulation — Global Importance of Rainforests

Rainforests are not merely local or regional ecosystems — they perform **critical services for the entire planet's climate system**. Their global importance extends far beyond the countries in which they are

physically located.

Carbon Storage and the Carbon Sink Function

Tropical rainforests together store approximately **250 billion tonnes of carbon** — roughly equivalent to 25 years of current global CO₂ emissions from all fossil fuel burning combined. This carbon is held in:

- **Living biomass** (above ground: trunks, branches, leaves; below ground: roots)
- **Dead wood** (fallen logs, bark, litter layer)
- **Soil organic matter** (particularly in peat-rich soils like those of Borneo)

When trees photosynthesise, they absorb CO₂ from the atmosphere and lock it into woody tissue — acting as a **carbon sink**. Intact tropical forests currently absorb a net 1.1–1.4 billion tonnes of carbon per year, significantly offsetting global emissions. However, this sink is weakening: a 2021 study in *Nature* found that heavily deforested and degraded parts of the Amazon (particularly in the south-east) have flipped to become net carbon **sources**, emitting more than they absorb. If this transition continues across the whole Amazon system, it would dramatically accelerate global warming.

Albedo and Surface Temperature Regulation

Rainforest canopies are relatively dark and absorb more solar radiation than bare or agricultural land. Counterintuitively, forest also causes significant **evaporative cooling** — the enormous volumes of water transpired by leaves absorb latent heat, cooling the local air temperature. Studies comparing deforested and forested areas in the tropics show that deforested land is on average **1–3°C warmer** than equivalent forested land. At a regional scale, large-scale deforestation warms regional temperatures, worsening drought conditions and potentially affecting atmospheric circulation patterns.

Influence on Global Weather Patterns

The Amazon acts as a major driver of atmospheric circulation. The enormous amounts of water vapour and heat released by evapotranspiration drive convection currents that influence the **position and strength of the jet stream** and inter-tropical weather systems. Climate modelling studies suggest that complete loss of the Amazon would significantly alter rainfall patterns as far away as the **Midwestern United States and parts of Europe** — demonstrating that the rainforest's climate services are genuinely global in reach, not merely regional.

Biodiversity as a Climate Stabiliser

High biodiversity itself contributes to ecosystem resilience and stability. A more diverse forest is better able to withstand and recover from disturbances such as drought, disease, or insect outbreaks. As climate change intensifies, maintaining high biodiversity in the rainforest may be critical to its ability to continue performing its climate regulation functions — a further reason why biodiversity loss and climate change are interlinked crises, not separate problems.

Carbon Sink: A natural system that absorbs more carbon (as CO₂) from the atmosphere than it releases. Intact tropical rainforests are major carbon sinks. When deforested, they become carbon sources — releasing stored carbon as CO₂.

Ecosystem Services: The benefits that healthy ecosystems provide to people, free of charge. For rainforests, key services include: carbon storage (climate regulation), water cycle regulation, biodiversity (gene pool for medicine and agriculture), local rainfall generation, and soil protection. These services are estimated to be worth trillions of dollars globally if costed — yet they are provided by nature without payment.

Exam Tip — Why Should Wealthy Countries Care? A common exam question asks why people in the UK should care about deforestation in Brazil or Indonesia. Key arguments: (1) CO₂ released anywhere affects global climate equally; (2) UK consumers drive demand for beef, soya, and palm oil that cause deforestation; (3) Rainforest biodiversity may hold future medical cures; (4) atmospheric circulation means Amazonian deforestation may affect European weather. This demonstrates understanding of globalisation, interdependence, and global responsibility.

12. A Small-Scale Ecosystem in the UK — Kenfig Dunes, South Wales

The specification requires one **small-scale UK ecosystem** studied in detail: its characteristics, the benefits it brings local communities, the challenges of managing it, and how different stakeholders value it. Kenfig National Nature Reserve, on the South Wales coast between Porthcawl and Port Talbot, is used here.

Characteristics

- **Size and setting:** Around 1,300 acres (roughly 520 hectares) of sand dunes, dune grassland, dune slacks and an open freshwater pool, Kenfig Pool, behind a wide beach. It is one of the largest surviving dune systems in the UK and is a National Nature Reserve.
- **A succession, not a single habitat:** Moving inland from the beach, conditions change and so does the vegetation — from bare, mobile sand at the strandline, through marram-grass-covered yellow dunes, to stabilised, humus-rich "grey" dunes and damp hollows called **slacks** where the water table reaches the surface. Each stage supports different species.
- **Abiotic controls:** Salt spray, a shortage of fresh water, high daytime surface temperatures on bare sand, sand blasting by onshore winds, and very low soil nutrient levels near the sea. Marram grass copes by rolling its leaves, growing waxy surfaces, and sending roots several metres down to reach moisture; it also grows upwards through accumulating sand, which is what builds a dune in the first place.
- **Biotic interest:** The slacks support a nationally significant population of the very rare **fen orchid** — a plant that needs open, damp, low-nutrient ground and disappears once taller vegetation shades it out. Numbers at Kenfig have recovered from around 200 plants to more than 4,000 following targeted management.

Dune slack: A damp hollow between dune ridges where the ground surface is close to, or seasonally below, the water table. Slacks are the most species-rich part of a dune system and the habitat on which Kenfig's rarest plants depend.

Benefits for Local Communities

- **Coastal protection:** The dune ridge absorbs wave energy and storm surge, protecting land, roads and property behind it — a natural sea defence maintained at no cost.
- **Recreation and wellbeing:** Free public access for walking, birdwatching, running and beach use, within easy reach of Bridgend, Porthcawl and Port Talbot.
- **Education and research:** Heavily used by schools and universities for fieldwork on succession, exactly the kind of study Component 3 assesses.
- **Local economy:** Visitors support nearby cafés, shops and accommodation in Porthcawl.
- **Regulating services:** The dune grassland and slack soils store carbon and filter groundwater.

The Management Challenge

The central problem at Kenfig is counter-intuitive: the dune system is becoming **too stable**. Historically, wind kept patches of sand bare and mobile, constantly resetting the succession and creating the open, low-nutrient conditions that rare species need. Over recent decades the proportion of bare sand has fallen sharply, and scrub and coarse grasses have spread. The causes are cumulative:

- The loss of traditional grazing by rabbits (after myxomatosis) and by livestock, so vegetation is no longer cropped short.
- Nitrogen deposition from vehicle and industrial emissions, which fertilises naturally poor sand and favours vigorous coarse plants over specialists.
- The spread of invasive and aggressive species, including sea buckthorn and dewberry.
- Trampling and informal path creation by visitors, which is damaging in the wrong place but, oddly, helpful where it keeps sand open.

Case Study — Managing Kenfig National Nature Reserve

Kenfig is owned by the **Kenfig Corporation Trust** and managed with **Natural Resources Wales (NRW)** under a five-year management agreement. Work carried out through NRW's **Sands of LIFE** dune restoration project includes:

- **Scrub clearance and turf stripping** to remove built-up vegetation and nutrient-rich topsoil, re-exposing bare sand and damp slack surfaces for species such as the fen orchid.
- **Re-introducing grazing**. A new 5 km livestock fence was agreed to allow an additional 200 hectares of dune grassland to be grazed sustainably, using animals to do the job rabbits once did.
- **Visitor management** — boardwalks, marked paths and a visitor centre that concentrate footfall away from the most sensitive slacks.

Outcome: The fen orchid population has risen from around 200 plants to over 4,000, which is the clearest single measure that intervention has worked.

Stakeholders and Conflicting Values

Stakeholder	How they value the ecosystem	Where conflict arises
Natural Resources Wales	As a protected habitat for rare species; success is measured in orchid counts and bare sand area	Clearing scrub and stripping turf looks destructive to visitors who expect conservation to mean "leave it alone"

Kenfig Corporation Trust / local residents	As common land with long-standing local rights and an important part of local identity	Fencing for grazing restricts free movement across land people have always walked
Visitors and dog walkers	As open space for recreation	Off-path walking and dogs disturb ground-nesting birds and trample slack vegetation; grazing livestock and loose dogs do not mix
Ecologists and schools	As a scientific and teaching resource	Repeated fieldwork on the same transects itself causes trampling
Nearby businesses	As a visitor attraction supporting trade	Want more access and facilities; managers want visitor numbers concentrated and controlled

Exam Tip — Small-Scale Ecosystem Questions: Eduqas wants a named, local, small ecosystem, not a biome. Learn three or four specific facts you can quote — location, size, one rare species, one management action, one measurable outcome — and one clear stakeholder conflict. A question on "the challenges of managing a small-scale ecosystem" is really asking you to explain why one group's idea of a successful outcome is another group's idea of damage.